

MODEL comparision matrix

Classification	Accuracy	Precision	Recall	F1Score	ROC/AUC
Logistic Regression	0.7171	0.1286	0.8158	0.2222	0.825
Decision Tree Classifier	0.7027	0.1074	0.6842	0.1857	0.7714
Random Forest Classifier	0.9166	0.1389	0.1316	0.1351	0.762
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.8592	0.15	0.3947	0.2174	0.7708
K-Nearest Neighbors Classifier	0.837	0.1009	0.2895	0.1497	0.6225
Support Vector Classifier (SVC)	0.8018	0.125	0.5	0.2	0.7406
Voting Vs Avearge (best 3 out of 6)	0.7471	0.1214	0.6579	0.2049	0.8084
Ensemble	0.8083	0.1438	0.5789	0.2304	0.7908

Which model gave you the best result and which was the worst?

Logistic Regression delivered the best overall results for our clinical diagnostic task. While its raw structural accuracy was lower (71.71%), it achieved the peak test class separation capability with an ROC/AUC score of 0.8250. Crucially, it minimized life-threatening false negatives by successfully identifying 81.58% of actual stroke events (highest overall Recall), which is the primary operational safety goal in medical screening software. The worst results were K-NN and Random Forest Classifier, performing the poorest. K-NN generated the lowest spatial classification boundary separation with a bottom ROC/AUC score of 0.6225. Meanwhile, Random Forest completely collapsed under majority class bias; despite applying over-sampling techniques, it generated a deceptive raw accuracy of 91.66% by guessing the safe non-stroke class, effectively missing 86.84% of active stroke patients with a catastrophic minority Rcall score of 13.16%.

Regression	MAE	MSE	Rsquared
Linear Regression	N/A	N/A	N/A
Decision Tree Regressor	N/A	N/A	N/A
Random Forest Regressor	N/A	N/A	N/A
Gradient Boosting Regressor (e.g., XGBoost or LightGBM)	N/A	N/A	N/A
K-Nearest Neighbors Regressor	N/A	N/A	N/A
Voting Vs Avearge (best 3 out of 6)	N/A	N/A	N/A
Ensemble	N/A	N/A	N/A

Which model gave you the best result and which was the worst?

N/A